

B6 Condensed-Matter Physics — Model Solutions, Trinity 2023

Paper A13485W1

The candidate is required to attempt two questions out of four. Below, full solutions to all four questions are provided in the Oxford lecturer style.

Question 1 — Neutron diffraction from NaTaO₃

Problem statement. The compound NaTaO₃ crystallises in a tetragonal lattice with $a = 0.555 \text{ nm}$, $c = 0.393 \text{ nm}$. The two-formula-unit cell contains two distinct oxygen sites with the fractional coordinates listed in the question; the four O₂ atoms are tilted by a small distortion parameter $\delta \ll 1$. We are asked to (a) derive the systematic-absence rule and the structure factor of Na, Ta, O₁ when $h + k = 2n$; (b) expand the O₂ contribution to first order in δ for various parities of h, k ; (c) compute, for a powder neutron experiment with $\lambda = 0.159 \text{ nm}$ and resolution 0.05° in 2θ , the squared structure factor, multiplicity and Bragg angle for the (001), (110) and (210) reflections; (d) deduce δ from the measured intensities $I_{110} + I_{001} = 1843$ and $I_{210} = 345$.

(a) Systematic absences from Na, Ta, O₁

The structure factor of an ion species X occupying N_X sites in the unit cell is

$$F_X(hkl) = b_X \sum_{j=1}^{N_X} \exp[2\pi i(hx_j + ky_j + lz_j)], \quad (1)$$

where (x_j, y_j, z_j) are the fractional coordinates of the j th atom of that species and b_X is the neutron coherent scattering length. Because the ions appear in pairs related by the centring vector $(\frac{1}{2}, \frac{1}{2}, 0)$, the four sets of coordinates can be summed in pairs and a common phase $\exp[i\pi(h+k)]$ factored out.

For Na at $(0, \frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{2}, 0, \frac{1}{2})$,

$$F_{\text{Na}} = b_{\text{Na}} e^{i\pi l} [e^{i\pi k} + e^{i\pi h}]. \quad (2)$$

For Ta at $(0, 0, 0)$ and $(\frac{1}{2}, \frac{1}{2}, 0)$,

$$F_{\text{Ta}} = b_{\text{Ta}} [1 + e^{i\pi(h+k)}]. \quad (3)$$

For O₁ at $(0, 0, \frac{1}{2})$ and $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$,

$$F_{\text{O}_1} = b_{\text{O}} e^{i\pi l} [1 + e^{i\pi(h+k)}]. \quad (4)$$

If $h + k$ is odd, then $e^{i\pi(h+k)} = -1$, so equations (3) and (4) vanish identically. Equation (2) is also zero, because $e^{i\pi k} = -e^{i\pi h}$ when h and k have opposite parity. Hence *none* of Na, Ta, O₁ contributes to a Bragg peak with $h + k = 2n + 1$. This is the centring-rule absence of the body-centred sublattice that these three ions populate.

If $h + k = 2n$ then $e^{i\pi h} = e^{i\pi k} = (-1)^h$, $e^{i\pi(h+k)} = +1$, and the three contributions sum to

$$F_{\text{Na+Ta+O}_1}(hkl) = 2[b_{\text{Ta}} + b_{\text{O}}(-1)^l + b_{\text{Na}}(-1)^{h+l}] \quad (h + k = 2n). \quad (5)$$

The pre-factor 2 counts the two formula units in the cell, and the alternating signs reflect the relative axial offsets of Na ($z = \frac{1}{2}$, plus a body-centring shift) and O₁ ($z = \frac{1}{2}$).

(b) Structure factor of O₂ to first order in δ

The four O₂ ions all lie in the $z = 0$ plane, so the l -dependent phase $e^{2\pi i l z}$ is unity for every term. Writing the four positions as

$$(\frac{1}{4} + \delta, -\frac{1}{4} + \delta, 0), (-\frac{1}{4} - \delta, \frac{1}{4} - \delta, 0), (\frac{1}{4} - \delta, \frac{1}{4} + \delta, 0), (-\frac{1}{4} + \delta, -\frac{1}{4} - \delta, 0),$$

the corresponding phases factorise in a useful way: terms 1 and 2 are complex conjugates, and so are terms 3 and 4. One finds

$$F_{O2} = 2b_O \left\{ \cos\left[\frac{\pi}{2}(h-k) + 2\pi\delta(h+k)\right] + \cos\left[\frac{\pi}{2}(h+k) + 2\pi\delta(k-h)\right] \right\}. \quad (6)$$

This expression is exact in δ . Expanding to first order using $\cos(\theta + \epsilon) \approx \cos\theta - \epsilon \sin\theta$,

$$\frac{F_{O2}}{2b_O} \approx \cos\frac{\pi(h-k)}{2} + \cos\frac{\pi(h+k)}{2} - 2\pi\delta \left[(h+k) \sin\frac{\pi(h-k)}{2} + (k-h) \sin\frac{\pi(h+k)}{2} \right]. \quad (7)$$

We now read off the three parity cases.

(i) h and k both odd. Both $(h \pm k)/2$ are integers, so the two sines vanish. The two cosines are $(-1)^{(h-k)/2}$ and $(-1)^{(h+k)/2}$; their indices differ by k (odd) and so have opposite parity. Therefore

$$\boxed{F_{O2} = 0 \quad (h, k \text{ both odd}).} \quad (8)$$

The two oxygen pairs contribute with opposite signs and cancel *exactly* (not merely to first order) for these reflections.

(ii) h and k both even. Again the sines vanish, and now $(h-k)/2$ and $(h+k)/2$ have the same parity (their difference is k , an even integer). The two cosines are equal, so

$$\boxed{F_{O2} = 4b_O (-1)^{(h+k)/2} \quad (h, k \text{ both even}).} \quad (9)$$

(iii) One of h, k even, the other odd. Now $(h \pm k)/2$ are half-integers, so both cosines vanish; only the δ -linear term survives. Writing $s_1 = (-1)^{(h-k-1)/2}$ and using $s_2/s_1 = (-1)^k$:

- If k is even (so h odd): $s_2 = s_1$, and (7) gives $F_{O2} = -2b_O \cdot 4\pi\delta s_1 k \cdot 0 \dots$ let us be careful. The combination is $[(h+k) + (k-h)] s_1 = 2k s_1$, hence

$$F_{O2} = -8\pi\delta b_O k s_1. \quad (10)$$

- If k is odd (so h even): $s_2 = -s_1$, giving

$$F_{O2} = -8\pi\delta b_O h s_1. \quad (11)$$

In both cases the magnitude scales as $|\delta| \times (\text{the odd index})$, so these "mixed-parity" reflections appear *only* because of the small O₂ tilt. They are the diagnostic peaks for measuring δ .

(c) Powder diffraction at $\lambda = 0.159$ nm

Lattice spacings. Using the orthogonal-axes formula $d^{-2} = (h^2 + k^2)/a^2 + l^2/c^2$:

$$d_{001} = c = 0.393 \text{ nm}, \quad (12)$$

$$d_{110} = a/\sqrt{2} = 0.555 \text{ nm}/\sqrt{2} = 0.39245 \text{ nm}, \quad (13)$$

$$d_{210} = a/\sqrt{5} = 0.555 \text{ nm}/\sqrt{5} = 0.24820 \text{ nm}. \quad (14)$$

Bragg angles. Solving $\sin \theta = \lambda/2d$:

$$2\theta_{001} = 2 \arcsin(0.159/0.786) = 2 \arcsin(0.20229) = 23.34^\circ, \quad (15)$$

$$2\theta_{110} = 2 \arcsin(0.159/0.7849) = 2 \arcsin(0.20257) = 23.38^\circ, \quad (16)$$

$$2\theta_{210} = 2 \arcsin(0.159/0.4964) = 2 \arcsin(0.32034) = 37.36^\circ. \quad (17)$$

The separation $|2\theta_{110} - 2\theta_{001}| \approx 0.04^\circ$ is below the experimental resolution of 0.05° , so these two reflections merge into a single peak — as advertised.

Structure factors. We use (5) together with the parity rules of part (b).

- (001): $h + k = 0$ (even), h, k both even. Equation (5) with $h = k = 0, l = 1$:

$$F_{\text{Na+Ta+O1}} = 2[b_{\text{Ta}} - b_{\text{O}} - b_{\text{Na}}] = 2(6.91 - 5.803 - 3.63) = -5.046 \text{ fm}. \quad (18)$$

Equation (9) with $(h + k)/2 = 0$ gives $F_{\text{O2}} = 4b_{\text{O}} = 23.212 \text{ fm}$. Hence

$$F(001) = -5.046 + 23.212 = 18.166 \text{ fm}, \quad |F(001)|^2 \approx 330 \text{ fm}^2. \quad (19)$$

- (110): $h + k = 2$ (even), h, k both odd. Using (5) with $h = k = 1, l = 0$,

$$F_{\text{Na+Ta+O1}} = 2[b_{\text{Ta}} + b_{\text{O}} - b_{\text{Na}}] = 2(6.91 + 5.803 - 3.63) = 18.166 \text{ fm}, \quad (20)$$

and (8) gives $F_{\text{O2}} = 0$. Therefore

$$|F(110)|^2 = (18.166 \text{ fm})^2 \approx 330 \text{ fm}^2. \quad (21)$$

- (210): $h + k = 3$ (odd), so by part (a) the Na+Ta+O₁ contribution vanishes. Only F_{O2} survives. Here $h = 2$ even, $k = 1$ odd, so equation (10) applies with $s_1 = (-1)^{(h-k-1)/2} = (-1)^0 = +1$:

$$F(210) = -8\pi\delta b_{\text{O}} k s_1 = -8\pi\delta b_{\text{O}} = -8\pi\delta \cdot 5.803 \text{ fm}, \quad (22)$$

$$|F(210)|^2 = 64\pi^2\delta^2 b_{\text{O}}^2 = 64\pi^2\delta^2 \cdot 33.675 \text{ fm}^2 \approx 2.128 \times 10^4 \delta^2 \text{ fm}^2. \quad (23)$$

Multiplicities for tetragonal symmetry. The tetragonal point group $4/mmm$ generates the following equivalents:

- (00 l) family: only (00 $\bar{l}$) is symmetry-related; multiplicity $m_{001} = 2$.
- ($hh0$) family with $h \neq 0$: the four members ($\pm h, \pm h, 0$); multiplicity $m_{110} = 4$.
- ($hk0$) with $h \neq k \neq 0$: the eight members ($\pm h, \pm k, 0$) and ($\pm k, \pm h, 0$); multiplicity $m_{210} = 8$.

Collecting:

(hkl)	$ F ^2$ (fm ²)	multiplicity	2θ
(001)	330	2	23.34°
(110)	330	4	23.38°
(210)	$2.13 \times 10^4 \delta^2$	8	37.36°

Since the (001) and (110) peaks are separated by only 0.04° and the resolution is 0.05°, they cannot be resolved and appear as a single observed peak with combined intensity $I_{001} + I_{110}$.

(d) Determination of δ

The integrated powder intensity is $I_{hkl} = K m_{hkl} |F_{hkl}|^2$ with the same arbitrary scale factor K for every reflection (geometrical Lorentz–polarisation factors having already been removed). Because $|F(001)|^2 = |F(110)|^2$, the merged peak satisfies

$$I_{001} + I_{110} = K (m_{001} + m_{110}) |F(110)|^2 = K \cdot 6 \cdot 330 \text{ fm}^2 = 1980 K \text{ fm}^2. \quad (24)$$

Setting this equal to the measured value 1843 fixes the scale,

$$K = 1843/1980 \text{ fm}^{-2} \approx 0.9308 \text{ fm}^{-2}. \quad (25)$$

(In other words, the unit of intensity is $K^{-1} \approx 1.074 \text{ fm}^2$, but we shall not need this conversion separately.)

For the (210) peak,

$$I_{210} = K m_{210} |F(210)|^2 = K \cdot 8 \cdot 64\pi^2 \delta^2 b_O^2 = 512\pi^2 K b_O^2 \delta^2. \quad (26)$$

Inserting $b_O^2 = 33.675 \text{ fm}^2$, $K = 0.9308$ and the measured $I_{210} = 345$:

$$\delta^2 = \frac{345}{512\pi^2 \cdot 0.9308 \cdot 33.675} = \frac{345}{1.583 \times 10^5} = 2.18 \times 10^{-3}. \quad (27)$$

Hence

$$\boxed{\delta \approx 0.047.} \quad (28)$$

This is small enough ($\delta \ll 1$) for the linearisation (7) to be self-consistent, and corresponds to an oxygen displacement of $\delta a \approx 2.6 \times 10^{-2} \text{ nm} \approx 0.26 \text{ \AA}$ — a typical octahedral-tilt amplitude in perovskite-related compounds.

Question 2 — Molecular orbital theory of H_2

Problem statement. Two identical atoms (sites 1 and 2) carry single-particle orbitals $|1\rangle, |2\rangle$ of energy ϵ_0 , with $(\hat{K} + \hat{V}_i) |i\rangle = \epsilon_0 |i\rangle$. A trial wavefunction $|\psi\rangle = c_1 |1\rangle + c_2 |2\rangle$ is used in the variational quotient $\lambda = \langle \psi | \hat{H} | \psi \rangle / \langle \psi | \psi \rangle$ for the two-atom Hamiltonian $\hat{H} = \hat{K} + \hat{V}_1 + \hat{V}_2$. The four matrix elements $V_{c1} = \langle 2 | \hat{V}_1 | 2 \rangle$, $V_{c2} = \langle 1 | \hat{V}_2 | 1 \rangle$, $t = -\langle 1 | \hat{V}_2 | 2 \rangle$ and $\sigma = \langle 1 | 2 \rangle$ are not assumed small, and overlap is retained.

(a) Secular equation without zero-overlap

The numerator and denominator of λ are

$$N(c_1, c_2) \equiv \langle \psi | \hat{H} | \psi \rangle = \sum_{ij} c_i^* c_j H_{ij}, \quad H_{ij} \equiv \langle i | \hat{H} | j \rangle, \quad (29)$$

$$D(c_1, c_2) \equiv \langle \psi | \psi \rangle = \sum_{ij} c_i^* c_j S_{ij}, \quad S_{ij} \equiv \langle i | j \rangle. \quad (30)$$

We compute the four matrix elements explicitly. For the diagonal entries, use $(\hat{K} + \hat{V}_1)|1\rangle = \epsilon_0|1\rangle$ to write $\hat{H}|1\rangle = \epsilon_0|1\rangle + \hat{V}_2|1\rangle$, hence

$$H_{11} = \langle 1|\hat{H}|1\rangle = \epsilon_0 + \langle 1|\hat{V}_2|1\rangle = \epsilon_0 + V_{c2}, \quad H_{22} = \epsilon_0 + V_{c1}. \quad (31)$$

For the off-diagonal entries, the same trick gives $\hat{H}|2\rangle = (\hat{K} + \hat{V}_2)|2\rangle + \hat{V}_1|2\rangle = \epsilon_0|2\rangle + \hat{V}_1|2\rangle$, so

$$H_{12} = \langle 1|\hat{H}|2\rangle = \epsilon_0\sigma + \langle 1|\hat{V}_1|2\rangle. \quad (32)$$

The remaining integral is converted via Hermiticity: $\langle 1|\hat{V}_1|2\rangle = \langle 2|\hat{V}_1|1\rangle^*$, and writing $\hat{H}|1\rangle = \epsilon_0|1\rangle + \hat{V}_2|1\rangle$ gives $\langle 2|\hat{H}|1\rangle = \epsilon_0\sigma^* + \langle 2|\hat{V}_2|1\rangle = \epsilon_0\sigma^* - t^*$. Therefore $H_{12} = \epsilon_0\sigma - t$, and similarly $H_{21} = \epsilon_0\sigma^* - t^*$. In matrix form

$$\mathbf{H} = \begin{pmatrix} \epsilon_0 + V_{c2} & \epsilon_0\sigma - t \\ \epsilon_0\sigma^* - t^* & \epsilon_0 + V_{c1} \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 1 & \sigma \\ \sigma^* & 1 \end{pmatrix}. \quad (33)$$

The condition for an extremum of $\lambda = N/D$ is $D \partial N / \partial c_i^* - N \partial D / \partial c_i^* = 0$, i.e. at the extremum

$$\frac{\partial N}{\partial c_i^*} = \lambda \frac{\partial D}{\partial c_i^*} \iff \sum_j H_{ij} c_j = \lambda \sum_j S_{ij} c_j. \quad (34)$$

Here we used the formal independence of c_i and c_i^* (equivalently, varying real and imaginary parts independently). Putting this in matrix notation and rearranging,

$$\boxed{(\mathbf{H} - \lambda \mathbf{S}) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0,} \quad (35)$$

which is precisely the matrix equation in the question. No assumption $\langle i|j\rangle = \delta_{ij}$ has been used.

(b) Symmetric two-atom case

Mirror symmetry $x \rightarrow -x$ exchanges the two atoms, $|1\rangle \leftrightarrow |2\rangle$ and $\hat{V}_1 \leftrightarrow \hat{V}_2$. Equating quantities related by this exchange,

$$V_{c1} = \langle 2|\hat{V}_1|2\rangle \longleftrightarrow \langle 1|\hat{V}_2|1\rangle = V_{c2} \Rightarrow V_{c1} = V_{c2} \equiv V_c.$$

Reality of σ and t : under $x \rightarrow -x$ the orbitals can be chosen with real spatial parts (1s-like, even-parity ground states), and the operators $\hat{V}_1, \hat{V}_2$ are real. Their integrals over real wavefunctions are real, so $\sigma, t \in \mathbb{R}$.

With $V_{c1} = V_{c2} = V_c$, t, σ real, the secular matrix becomes

$$\begin{pmatrix} \epsilon_0 + V_c - \lambda & (\epsilon_0 - \lambda)\sigma - t \\ (\epsilon_0 - \lambda)\sigma - t & \epsilon_0 + V_c - \lambda \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0. \quad (36)$$

Diagonal entries are equal and so are off-diagonal entries, so the eigenvectors are the symmetric and antisymmetric combinations $c_1 = \pm c_2$. Substituting back gives the eigenvalues

$$\lambda_+ (c_1 = +c_2, \text{ bonding}) : (1 + \sigma)\lambda_+ = \epsilon_0(1 + \sigma) + V_c - t \Rightarrow \boxed{\lambda_+ = \epsilon_0 + \frac{V_c - t}{1 + \sigma}} \quad (37)$$

$$\lambda_- (c_1 = -c_2, \text{ antibonding}) : (1 - \sigma)\lambda_- = \epsilon_0(1 - \sigma) + V_c + t \Rightarrow \boxed{\lambda_- = \epsilon_0 + \frac{V_c + t}{1 - \sigma}} \quad (38)$$

The corresponding orthonormal eigenstates are

$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2(1 + \sigma)}}, \quad |\psi_-\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2(1 - \sigma)}}. \quad (39)$$

The norm factors come from $\langle \psi_\pm | \psi_\pm \rangle = 2(1 \pm \sigma)/[2(1 \pm \sigma)] = 1$; without retaining σ one would mis-normalise the bonding orbital.

(c) Hydrogen molecule with explicit integrals

Both t and σ involve overlaps of the type $\int \phi(\mathbf{r})f(\mathbf{r})\phi(\mathbf{r} - \mathbf{d})d^3r$. The hydrogen ground-state $\phi(r) = \pi^{-1/2}e^{-r}$ decays exponentially, so as $d \rightarrow \infty$ the integrand is concentrated only in the narrow region between the two centres and contains the product $e^{-r_1}e^{-r_2}$, yielding an exponentially small result. By contrast, $V_c = \langle 1 | \hat{V}_2 | 1 \rangle$ is the electrostatic energy of the charge density $|\phi_1|^2$ in the Coulomb potential of the distant proton; once $d \gg a_0$, $|\phi_1|^2$ has unit total charge and looks like a point at the origin from far away, so $V_c \rightarrow -2/d$ (in Rydberg units), the bare Coulomb potential of one proton seen by an electron of unit charge density at the other.

(ii) Ground-state energy at $d_{\text{exp}} = 1.4$. Inserting $d = 1.4$ into the given expressions:

$$t = 2e^{-1.4}(1 + 1.4) = 2 \cdot 0.2466 \cdot 2.4 = 1.184, \quad (40)$$

$$\sigma = e^{-1.4}(1 + 1.4 + 1.4^2/3) = 0.2466 \cdot 3.0533 = 0.7530, \quad (41)$$

$$V_c = \frac{2[e^{-2.8}(1 + 1.4) - 1]}{1.4} = \frac{2(0.06081 \cdot 2.4 - 1)}{1.4} = \frac{2(-0.8541)}{1.4} = -1.220. \quad (42)$$

The bonding eigenvalue (37) (with $\epsilon_0 = -1 E_R$) is

$$\lambda_+^{(\sigma)} = -1 + \frac{V_c - t}{1 + \sigma} = -1 + \frac{-1.220 - 1.184}{1.7530} = -1 - 1.371 = -2.371 E_R. \quad (43)$$

Neglecting overlap ($\sigma \rightarrow 0$) instead would give

$$\lambda_+^{(\sigma=0)} = -1 + (V_c - t) = -1 - 2.404 = -3.404 E_R. \quad (44)$$

The 2-electron MO ground state places both electrons in $|\psi_+\rangle$ with opposite spin, so the electronic part of the molecular energy is $2\lambda_+$. Adding the proton-proton repulsion $2/d$ (Rydberg units),

$$E_{\text{tot}}^{(\sigma)} = 2(-2.371) + 2/1.4 = -4.742 + 1.4286 = -3.314 E_R, \quad (45)$$

$$E_{\text{tot}}^{(\sigma=0)} = 2(-3.404) + 2/1.4 = -6.807 + 1.4286 = -5.379 E_R. \quad (46)$$

Comparing to the dissociation limit $E_{2H} = 2\epsilon_0 = -2 E_R$, the predicted dissociation energies are $E_d^{(\sigma)} = 2 - 3.314 = 1.31 E_R$ and $E_d^{(\sigma=0)} = 3.38 E_R$. The two values differ by a factor of three, so the overlap σ *cannot* be neglected: it both renormalises the bonding orbital and reduces the apparent binding energy substantially.

(iii) Comparison with experiment. The observed dissociation energy is $E_d^{\text{exp}} \approx 0.33 E_R$, an order of magnitude smaller than (43) predicts. The MO answer therefore overbinds the molecule by roughly $1 E_R \approx 13.6 \text{ eV}$.

(iv) Behaviour at $d = 0$. Take $d \rightarrow 0$ in the given expressions: $t \rightarrow 2(1 + 0) = 2$, $\sigma \rightarrow 1$, $V_c \rightarrow \lim_{d \rightarrow 0} 2[e^{-2d}(1 + d) - 1]/d = 2 \frac{d}{dd} [e^{-2d}(1 + d)]_0 = 2(-2 + 1) = -2$. Then

$$\lambda_+(d \rightarrow 0) = -1 + \frac{-2 - 2}{1 + 1} = -1 - 2 = -3 E_R, \quad (47)$$

which is more negative than $-2.371 E_R$ at d_{exp} ; hence the MO eigenvalue λ_+ keeps decreasing as $d \rightarrow 0$ (and indeed has its absolute minimum at $d = 0$, as stated). Of course in the full molecular energy $2\lambda_+ + 2/d$ the proton repulsion term diverges, so $E_{\text{tot}} \rightarrow +\infty$ at $d = 0$ and there is still a bound minimum at finite d , but the eigenvalue λ_+ alone does not register the singularity.

(d) Discussion: which physics is missing

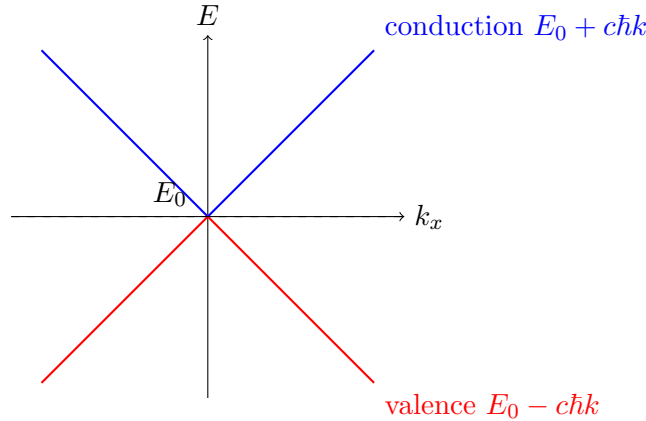
The simple MO calculation of part (c) overestimates the dissociation energy by a factor of ~ 4 at d_{exp} and predicts an unphysical minimum of λ_+ at $d = 0$. Both errors stem from a single omitted physical effect: *electron–electron repulsion*. The trial state $|\psi_+\rangle^2$ places *both* electrons in the same spatial orbital $|\psi_+\rangle$, with appreciable amplitude on each centre. When the molecule is squeezed to small d , the two electrons are forced into the same small region of space and their mutual Coulomb repulsion $\sim 1/r_{12}$ grows large. Including this Hartree term $J = \langle \psi_+ \psi_+ | 1/r_{12} | \psi_+ \psi_+ \rangle > 0$ raises the energy at small d , removes the spurious $d = 0$ minimum and weakens the binding at d_{exp} towards the experimental $0.33 E_R$. A second related deficiency is the absence of *correlation* between the two electrons: when the bond stretches, the simple MO state still has equal amplitude for both electrons on the same atom (ionic configurations $\text{H}^- \text{H}^+$) as for them on different atoms, with the result that dissociation is very poorly described. The Heitler–London (valence-bond) treatment, or configuration interaction with the antibonding orbital, repairs this defect. Both fixes act in the same direction: they lower the energy less at intermediate d and lower it more at large d , bringing the binding curve into agreement with experiment.

Question 3 — Massless Dirac fermions in 2D

Problem statement. A two-dimensional material has electronic dispersion $E = E_0 \pm c\hbar k$ near the Fermi energy, with $c = 1.0 \times 10^6 \text{ m/s}$. We are asked to (a) sketch the energy surfaces and contrast with a $E_g \rightarrow 0$ direct-gap semiconductor; (b) derive the carrier density in the conduction band as $n_{\text{el}} = C_1 \beta^p F_1(\alpha)$; (c) derive the electronic specific heat in the form $C_2 k_B^3 T^q [a F_2(\alpha) + b \alpha F_1(\alpha)]$; (d) at $\tilde{\mu} = 0$ relate electron and hole populations and use the small- α expansion of F_2 to obtain the total specific heat at 100 K; (e) deduce the limiting behaviour at $\tilde{\mu} \gg k_B T$, and the $T \rightarrow 0$ relation $\mu = E_0 + C_3 n_{\text{el}}^r$.

(a) Energy surfaces and comparison with $E_g \rightarrow 0$ semiconductor

Plotting E versus (k_x, k_y) , the conduction band is the upper cone $E = E_0 + c\hbar k$ and the valence band the lower cone $E = E_0 - c\hbar k$. The two cones meet at the single Dirac point $(k_x, k_y) = (0, 0)$. At $E_F = E_0$ the Fermi "surface" reduces to that point.



(ii) Comparison with a vanishing-gap parabolic semiconductor. Both systems have zero gap at the band-touching point and are therefore semimetallic. The crucial differences are:

- **Dispersion:** linear (massless) here vs. quadratic (parabolic) in a conventional $E_g \rightarrow 0$ semiconductor. Carriers in the present material have an energy-independent group velocity $|\mathbf{v}_g| = c$, exactly analogous to relativistic massless particles.
- **Density of states:** $g(E) \propto |E - E_0|$ here, vanishing linearly at the Dirac point, whereas the parabolic 2D semiconductor has $g \rightarrow \text{const}$ there.
- **Effective mass:** undefined (or "zero") for the linear band; finite for the parabolic case.
- **Topology of the wavefunction:** for the linear band the eigenspinors carry a Berry phase π around the Dirac point, leading to suppression of backscattering; the parabolic-band system does not.

(b) Conduction-band density

The number of single-particle states per unit area in k -space is $1/(2\pi)^2$, with an additional factor of 2 for spin. In a circular shell of radius k to $k + dk$ the area is $2\pi k dk$, so the density of states per unit area as a function of energy in the conduction band ($E \geq E_0$) is

$$g(E) = 2 \cdot \frac{2\pi k}{(2\pi)^2} \frac{dk}{dE} = \frac{k}{\pi} \cdot \frac{1}{c\hbar} = \frac{E - E_0}{\pi c^2 \hbar^2}. \quad (48)$$

Integrating against the Fermi function $f(E) = [e^{\beta(E-\mu)} + 1]^{-1}$ and substituting $u = \beta(E - E_0)$ ($du = \beta dE$),

$$\begin{aligned} n_{\text{el}} &= \int_{E_0}^{\infty} g(E) f(E) dE = \frac{1}{\pi c^2 \hbar^2} \int_0^{\infty} \frac{(u/\beta) du/\beta}{e^{u-\alpha} + 1} \\ &= \frac{1}{\pi c^2 \hbar^2 \beta^2} \int_0^{\infty} \frac{u du}{e^{u-\alpha} + 1} = \frac{1}{\pi c^2 \hbar^2 \beta^2} \cdot 1! \cdot F_1(\alpha), \end{aligned} \quad (49)$$

using the definition $F_j(x) = (1/j!) \int_0^{\infty} t^j dt / (e^{t-x} + 1)$. Therefore

$$\boxed{n_{\text{el}} = \frac{1}{\pi c^2 \hbar^2} \beta^{-2} F_1(\alpha), \quad C_1 = \frac{1}{\pi c^2 \hbar^2}, \quad p = -2.} \quad (50)$$

The T^2 behaviour mirrors the linear DOS: in 2D with a constant DOS one obtains T^0 , in 2D with a linear DOS one obtains T^2 .

(c) Conduction-band specific heat

The electronic energy density per unit area, measured from the band edge E_0 , is

$$\tilde{u}_{\text{el}} = \int_{E_0}^{\infty} (E - E_0) g(E) f(E) dE. \quad (51)$$

Using (48) and again $u = \beta(E - E_0)$,

$$\tilde{u}_{\text{el}} = \frac{1}{\pi c^2 \hbar^2 \beta^3} \int_0^{\infty} \frac{u^2 du}{e^{u-\alpha} + 1} = \frac{2}{\pi c^2 \hbar^2 \beta^3} F_2(\alpha) = \frac{2(k_B T)^3}{\pi c^2 \hbar^2} F_2(\alpha), \quad (52)$$

since $F_2(\alpha) = \frac{1}{2!} \int_0^{\infty} u^2 du / (e^{u-\alpha} + 1)$. At constant chemical potential μ (and hence constant $\tilde{\mu} = \mu - E_0$),

$$\left. \frac{d\alpha}{dT} \right|_{\mu} = \frac{d}{dT} (\tilde{\mu}/k_B T) = -\frac{\alpha}{T}. \quad (53)$$

Differentiating (52) and using $dF_2/d\alpha = F_1$,

$$\begin{aligned} c_v^{\text{el}} &= \left. \frac{d\tilde{u}_{\text{el}}}{dT} \right|_{\mu} = \frac{2}{\pi c^2 \hbar^2} \left[3k_{\text{B}}^3 T^2 F_2(\alpha) + (k_{\text{B}} T)^3 F_1(\alpha) \frac{d\alpha}{dT} \right] \\ &= \frac{2k_{\text{B}}^3 T^2}{\pi c^2 \hbar^2} [3F_2(\alpha) - \alpha F_1(\alpha)]. \end{aligned} \quad (54)$$

Hence

$$\boxed{c_v^{\text{el}} = \frac{2}{\pi c^2 \hbar^2} k_{\text{B}}^3 T^2 [3F_2(\alpha) - \alpha F_1(\alpha)], \quad C_2 = \frac{2}{\pi c^2 \hbar^2}, \quad q = 2, \quad a = 3, \quad b = -1.} \quad (55)$$

The factor αF_1 subtracts the energy that flows into changing the population through μ : at $\tilde{\mu} = 0$ it vanishes and the heat capacity is purely $\propto F_2$.

(d) Charge neutrality at $\tilde{\mu} = 0$ and total specific heat

(i) Equality of electron and hole numbers. The valence band $E = E_0 - \hbar k$ has DOS $g_v(E) = (E_0 - E)/(\pi c^2 \hbar^2)$ for $E \leq E_0$, identical in form to the conduction band reflected about E_0 . The hole occupation is $1 - f(E) = [e^{\beta(\mu - E)} + 1]^{-1}$. Substituting $u = \beta(E_0 - E)$,

$$n_{\text{h}} = \int_{-\infty}^{E_0} g_v(E) [1 - f(E)] dE = \frac{1}{\pi c^2 \hbar^2 \beta^2} \int_0^{\infty} \frac{u du}{e^{u+\alpha} + 1} = \frac{F_1(-\alpha)}{\pi c^2 \hbar^2 \beta^2}. \quad (56)$$

At $\tilde{\mu} = 0$ ($\alpha = 0$) we have $F_1(-0) = F_1(0)$, so $n_{\text{h}} = n_{\text{el}}$ identically. By the same reflection symmetry the hole specific heat is given by (55) with $\alpha \rightarrow -\alpha$, equal at $\alpha = 0$ to the electron contribution.

(ii) Total specific heat at $\tilde{\mu} = 0$. Combining electron and hole contributions and setting $\alpha = 0$,

$$c_v^{\text{tot}}|_{\tilde{\mu}=0} = 2 \cdot \frac{2k_{\text{B}}^3 T^2}{\pi c^2 \hbar^2} \cdot 3F_2(0) = \frac{12k_{\text{B}}^3 T^2}{\pi c^2 \hbar^2} F_2(0). \quad (57)$$

Using the small- α expansion $F_2(\alpha) \approx 3\zeta/4 + \pi^2\alpha/12 + \frac{1}{2}\alpha^2 \ln 2$ at $\alpha = 0$ gives $F_2(0) = 3\zeta/4$ with $\zeta = 1.2021$. Therefore

$$\boxed{c_v^{\text{tot}}|_{\tilde{\mu}=0} = \frac{12 \cdot 3\zeta/4}{\pi c^2 \hbar^2} k_{\text{B}}^3 T^2 = \frac{9\zeta k_{\text{B}}^3 T^2}{\pi c^2 \hbar^2}.} \quad (58)$$

Numerically, with $k_{\text{B}} = 1.381 \times 10^{-23}$ J/K, $\hbar = 1.055 \times 10^{-34}$ Js, $c = 1.0 \times 10^6$ m/s and $T = 100$ K:

$$\pi c^2 \hbar^2 = \pi \cdot 10^{12} \cdot (1.055 \times 10^{-34})^2 = 3.497 \times 10^{-56} \text{ J}^2 \text{ m}^2, \quad (59)$$

$$k_{\text{B}}^3 T^2 = (1.381 \times 10^{-23})^3 \cdot 10^4 = 2.634 \times 10^{-65} \text{ J}^3/\text{K}, \quad (60)$$

$$c_v^{\text{tot}} = \frac{9 \cdot 1.2021 \cdot 2.634 \times 10^{-65}}{3.497 \times 10^{-56}} \approx 8.15 \times 10^{-9} \text{ J K}^{-1} \text{ m}^{-2}. \quad (61)$$

This is the heat capacity per unit area of a single graphene-like layer at 100 K ($\sim 8 \text{ nJ K}^{-1} \text{ m}^{-2}$); per electron it is comparable to k_{B} but the small carrier density makes the total quite small.

(e) Degenerate limit $\tilde{\mu} \gg k_{\text{B}} T$

Taking the given approximation $F_2(\alpha) \approx \alpha^3/6 + \pi^2\alpha/6$ in the limit $\alpha \rightarrow \infty$, and using $F_1 = dF_2/d\alpha \approx \alpha^2/2 + \pi^2/6$:

(i) **Temperature dependence of n_{el} .** From (50)

$$n_{\text{el}} = \frac{(k_{\text{B}}T)^2}{\pi c^2 \hbar^2} \left(\frac{\alpha^2}{2} + \frac{\pi^2}{6} \right) = \frac{1}{\pi c^2 \hbar^2} \left(\frac{\tilde{\mu}^2}{2} + \frac{\pi^2}{6} (k_{\text{B}}T)^2 \right). \quad (62)$$

At leading order $n_{\text{el}} = \tilde{\mu}^2 / (2\pi c^2 \hbar^2)$, independent of T : the carrier density becomes the zero-temperature Fermi-volume value plus a Sommerfeld T^2 correction.

(ii) **Chemical potential at $T \rightarrow 0$.** Inverting the leading term of (62),

$$\tilde{\mu}^2 \rightarrow 2\pi c^2 \hbar^2 n_{\text{el}} \Rightarrow \mu \rightarrow E_0 + \hbar c \sqrt{2\pi n_{\text{el}}}. \quad (63)$$

Therefore

$$\boxed{C_3 = \hbar c \sqrt{2\pi}, \quad r = \frac{1}{2}.} \quad (64)$$

The half-power dependence is the 2D Dirac analogue of $\mu \propto n^{2/3}$ in the 3D parabolic case and $\mu \propto n$ in the 2D parabolic case — it follows directly from the linear DOS $g(E) \propto E$.

Question 4 — Easy-axis anisotropy with transverse field, $j = 1$

Problem statement. An ion of angular momentum j has Hamiltonian

$$\hat{H} = -D \hat{J}_x^2 - g\mu_{\text{B}} \hat{J}_z B, \quad D > 0,$$

with $\hat{J}_x^2 = \frac{1}{4}(\hat{J}_+^2 + \hat{J}_-^2) + \frac{1}{2}(\hat{J}^2 - \hat{J}_z^2)$. We are asked to (a) construct the matrix representation for $j = 1$ in the basis $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$, identifying a and b ; (b) diagonalise and expand to lowest order in B ; (c) compute the partition function and transverse susceptibility and identify $f_{\perp}(D\beta)$; (d) verify limits using the given $f_{\parallel}$; (e) interpret the figure and extract D .

(a) **Matrix representation in the $j = 1$ basis**

In the basis $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$:

- $\hat{J}^2 = 2\hbar^2 \mathbb{I}$ (we set $\hbar = 1$ for the matrix element coefficients).
- $\hat{J}_z = \text{diag}(1, 0, -1)$, so $\hat{J}_z^2 = \text{diag}(1, 0, 1)$.
- $\hat{J}^2 - \hat{J}_z^2 = \text{diag}(2 - 1, 2 - 0, 2 - 1) = \text{diag}(1, 2, 1)$.
- $\hat{J}_+ |1, -1\rangle = \sqrt{2} |1, 0\rangle$ and $\hat{J}_+ |1, 0\rangle = \sqrt{2} |1, 1\rangle$, so $\hat{J}_+^2 |1, -1\rangle = 2 |1, 1\rangle$ and $\hat{J}_+^2$ vanishes on the other two basis kets. By Hermitian conjugation $\hat{J}_-^2 |1, 1\rangle = 2 |1, -1\rangle$. Hence the matrix of $(\hat{J}_+^2 + \hat{J}_-^2)/4$ has only the (1,3) and (3,1) entries equal to 1/2.

Adding the diagonal contribution $\frac{1}{2}(\hat{J}^2 - \hat{J}_z^2) = \text{diag}(\frac{1}{2}, 1, \frac{1}{2})$ gives

$$\hat{J}_x^2 = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}, \quad (65)$$

so that the anisotropy term $-D \hat{J}_x^2 = -D \text{diag}[(\frac{1}{2}, 1, \frac{1}{2})]$ matches the form in the question with $\boxed{a = \frac{1}{2}}$.

The Zeeman term $-g\mu_{\text{B}} B \hat{J}_z = -g\mu_{\text{B}} B \text{diag}(1, 0, -1)$ then identifies $\boxed{b = 1}$.

(b) Eigenvalues to lowest order in B

The matrix $\mathbf{H}$ block-decomposes: the central column/row (the $|1,0\rangle$ subspace) is decoupled with eigenvalue $-D \cdot 2a = -D$ at all B . The remaining 2×2 block in the $\{|1,1\rangle, |1,-1\rangle\}$ subspace is

$$\mathbf{H}' = \begin{pmatrix} -D/2 - g\mu_B B & -D/2 \\ -D/2 & -D/2 + g\mu_B B \end{pmatrix}. \quad (66)$$

Its trace is $-D$ and its determinant is $(D^2/4) - (g\mu_B B)^2 - (D/2)^2 = -(g\mu_B B)^2$, so the eigenvalues are

$$E_{\pm} = -\frac{D}{2} \pm \frac{1}{2} \sqrt{D^2 + 4(g\mu_B B)^2}. \quad (67)$$

For $g\mu_B B \ll D$ expand the square root,

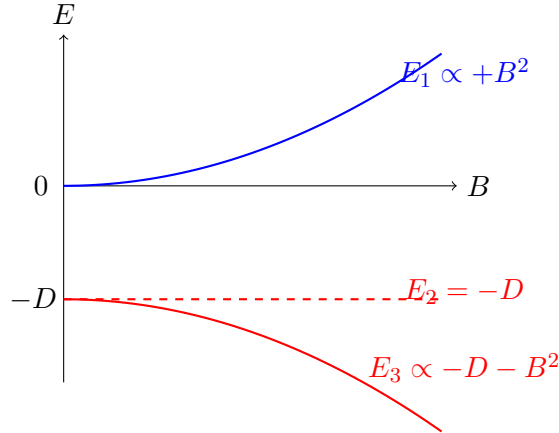
$$\sqrt{D^2 + 4(g\mu_B B)^2} \approx D + \frac{2(g\mu_B B)^2}{D}, \quad (68)$$

so that to second order in B ,

$$\boxed{E_1 = +\frac{(g\mu_B B)^2}{D}, \quad E_2 = -D, \quad E_3 = -D - \frac{(g\mu_B B)^2}{D}.} \quad (69)$$

At $B = 0$ the spectrum is $\{0, -D, -D\}$: a doublet at $-D$ (the two eigenstates of $\hat{J}_x$ with $m_x = \pm 1$) and a singlet at 0 (the $m_x = 0$ state). A transverse field couples the two degenerate $\hat{J}_z$ eigenstates and splits the doublet quadratically; the singlet $|1,0\rangle$ is unshifted to all orders in B because $\hat{J}_z |1,0\rangle = 0$.

Sketch. The three energies as functions of B :



(c) Partition function and low-field transverse susceptibility

(i) **Partition function.** For N non-interacting ions,

$$Z_N(B) = [Z_1(B)]^N, \quad Z_1 = e^{-\beta E_1} + e^{-\beta E_2} + e^{-\beta E_3}. \quad (70)$$

Inserting (69) and writing $w \equiv (g\mu_B)^2/D$,

$$Z_1(B) = e^{-\beta w B^2} + e^{\beta D} + e^{\beta D + \beta w B^2}. \quad (71)$$

(ii) **Magnetic moment in the low-field limit.** Expand to $O(B^2)$:

$$Z_1(B) \approx 1 + 2e^{\beta D} + \beta w B^2 (e^{\beta D} - 1). \quad (72)$$

The free energy per ion is $F_1 = -\beta^{-1} \ln Z_1$, so the moment per ion is

$$m_1 = -\frac{\partial F_1}{\partial B} = \frac{1}{\beta} \frac{\partial \ln Z_1}{\partial B} = \frac{2\beta w B (e^{\beta D} - 1)}{1 + 2e^{\beta D}} \cdot \frac{1}{\beta} = \frac{2w B (e^{\beta D} - 1)}{1 + 2e^{\beta D}} + O(B^3). \quad (73)$$

The total moment of the ensemble is $m = N m_1$.

(iii) **Transverse susceptibility.** By definition $\chi_{\perp} = \mu_0 dm/dB|_{B \rightarrow 0}$:

$$\chi_{\perp} = \mu_0 \cdot \frac{2N(g\mu_B)^2}{D} \cdot \frac{e^{\beta D} - 1}{1 + 2e^{\beta D}}. \quad (74)$$

Multiplying and dividing by $\beta D/3$ to make the Curie susceptibility $\chi_C = 2N\mu_0(g\mu_B)^2\beta/3$ explicit,

$$\chi_{\perp} = \chi_C \cdot \frac{3}{D\beta} \cdot \frac{e^{D\beta} - 1}{1 + 2e^{D\beta}}. \quad (75)$$

Hence

$$\boxed{\chi_{\perp} = \chi_C f_{\perp}(D\beta), \quad f_{\perp}(x) = \frac{3}{x} \cdot \frac{e^x - 1}{1 + 2e^x}.} \quad (76)$$

(d) **Limits and inequality with the parallel case**

The reference functions are $f_{\parallel}(x) = 3e^x/(1 + 2e^x)$ and (above) $f_{\perp}(x) = (3/x)(e^x - 1)/(1 + 2e^x)$.

(i) **Asymptotic $T \rightarrow \infty$ ($x \rightarrow 0$).** Use $e^x - 1 \approx x$ and $1 + 2e^x \approx 3$:

$$f_{\parallel}(0) = 3 \cdot 1/3 = 1, \quad (77)$$

$$f_{\perp}(0) = (3/x) \cdot x/3 = 1. \quad (78)$$

So $\chi_{\parallel}, \chi_{\perp} \rightarrow \chi_C$ at high temperature, as expected: the anisotropy energy D is negligible compared to thermal energy and the system behaves as a free $j = 1$ paramagnet.

(ii) $\chi_{\parallel} > \chi_C$ **at all T .** We need $f_{\parallel}(x) > 1$ for all $x > 0$:

$$\frac{3e^x}{1 + 2e^x} > 1 \iff 3e^x > 1 + 2e^x \iff e^x > 1, \quad (79)$$

which holds for all $x > 0$. Thus $\chi_{\parallel} > \chi_C$. Physically, when the field is along the easy axis, the lowest two states $|m_x = \pm 1\rangle$ are split linearly in B , so the population imbalance grows faster than for a free spin and the susceptibility is enhanced.

(iii) $\chi_{\perp}$ **finite at $T = 0$.** In the low-temperature limit $x \rightarrow \infty$, $e^x - 1 \sim e^x$ and $1 + 2e^x \sim 2e^x$, so

$$\chi_{\perp} \rightarrow 2N\mu_0(g\mu_B)^2/D \cdot \frac{1}{2} = \frac{N\mu_0(g\mu_B)^2}{D}. \quad (80)$$

This is finite — a Van Vleck-type response. Mechanism: with the field perpendicular to the easy axis, the ground states $|m_x = \pm 1\rangle$ have $\langle \hat{J}_z \rangle = 0$ but the field admixes them with the $|m_x = 0\rangle$ excited state at energy D above; second-order perturbation theory gives a B -linear induced moment whose coefficient is independent of temperature once $k_B T \ll D$. In contrast $\chi_C \propto 1/T$ diverges, so $f_{\perp}(x) = \chi_{\perp}/\chi_C \rightarrow 0$ at $T = 0$ even though $\chi_{\perp}$ itself does not.

(e) Interpretation of the figure and determination of D

Physical picture. The plot shows three molar susceptibilities for $g = 2$ as functions of temperature. At high T (> 25 K) all three converge to a single curve, the Curie tail, because $k_B T \gg D$ and the ion behaves isotropically. At low T the anisotropy splits the response: $\chi_{\parallel}$ is a factor $3/2$ larger than χ_C (because $f_{\parallel} \rightarrow 3/2$ as $x \rightarrow \infty$) but still diverges as $1/T$, while $\chi_{\perp}$ saturates at the Van Vleck value $N_A \mu_0 (g\mu_B)^2 / D$. This is the classic behaviour of a single-ion easy-axis anisotropy with the field along ($\parallel$) or transverse to ($\perp$) the anisotropy axis.

Extracting D . Reading off the saturation value of $\chi_{\perp}$ at $T \rightarrow 0$ from the figure, $\chi_{\perp}^{\text{sat}} \approx 3 \times 10^{-6} \text{ m}^3/\text{mol}$, and equating to the Van Vleck expression,

$$\frac{N_A \mu_0 (g\mu_B)^2}{D} = \chi_{\perp}^{\text{sat}} \Rightarrow D = \frac{N_A \mu_0 (g\mu_B)^2}{\chi_{\perp}^{\text{sat}}}. \quad (81)$$

With $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$, $\mu_0 = 1.2566 \times 10^{-6} \text{ H/m}$, $g = 2$, $\mu_B = 9.274 \times 10^{-24} \text{ J/T}$:

$$\begin{aligned} N_A \mu_0 (g\mu_B)^2 &= 6.022 \times 10^{23} \cdot 1.2566 \times 10^{-6} \cdot (1.855 \times 10^{-23})^2 \\ &= 6.022 \times 10^{23} \cdot 1.2566 \times 10^{-6} \cdot 3.441 \times 10^{-46} \\ &\approx 2.604 \times 10^{-28} \text{ J m}^3/\text{mol}, \end{aligned} \quad (82)$$

hence

$$D \approx \frac{2.604 \times 10^{-28} \text{ J m}^3/\text{mol}}{3 \times 10^{-6} \text{ m}^3/\text{mol}} = 8.7 \times 10^{-23} \text{ J} \approx k_B \cdot 6.3 \text{ K}. \quad (83)$$

Equivalently $D/k_B \approx 6 \text{ K}$, consistent with the temperature at which the figure shows the three curves diverging from each other.

$$\boxed{D \approx 8.7 \times 10^{-23} \text{ J} \quad (D/k_B \approx 6 \text{ K}).} \quad (84)$$

A useful internal consistency check: the curves visibly merge once $T \gtrsim 4D/k_B \approx 25 \text{ K}$, which matches the regime where $f_{\perp} \approx f_{\parallel} \approx 1$, exactly as found in part (d)(i).